CHAPTER #07

Cool the Case

Ensuring a cool, quiet, and stylish PC from the inside out

Your cooler pulls heat off your CPU/GPU. Your case moves that heat out and fresh air in. Treat them as a system: the right cooler in the wrong case (or vice versa) leaves performance on the table, raises noise, and shortens component life—especially in India's warmer months.

Cooling basics you should actually care about

- **Thermal throttling:** When parts get hot, they slow down to stay safe. Good cooling = stable turbo clocks, fewer stutters.
- **Airflow path:** Cool air in (front/bottom), hot air out (rear/top).
- **Positive pressure:** Slightly more intake than exhaust reduces dust entering through gaps.
- **Ambient limits:** Fans can't cool below room temperature; hot rooms = higher component temps.

The big three: air coolers, AIOs, custom loops

Use this quick read to find your fit.

Air coolers (heatsink + fan)

- **Pros:** Simple, reliable, affordable, easy to clean. Excellent for mainstream to high-end CPUs with a quality twin-tower.
- **Watch for:** Height clearance vs case; RAM clearance (tall heat spreaders); good case airflow.
- **Who it suits:** Most gaming and creator builds that value low fuss and long-term reliability.

AIO liquid coolers (sealed loop)

- **Pros:** Strong sustained cooling, tidy RAM/VRM access, clean aesthetics. 240/280 mm hits the sweet spot; 360 mm for top chips.